

Muhammad Waqar

AI Lead Engineer — Generative AI, RAG Systems, AI Agents

Islamabad, Pakistan | waqarsahi621@gmail.com | +92 308 6715563 | linkedin.com/in/muhammad-waqar-1a594411a |
github.com/MuhammadWaqar621 | muhammadwaqar-ai.vercel.app

SUMMARY

AI Lead Engineer with 7+ years of experience building production AI systems, including RAG pipelines, multi-agent workflows, and computer vision. Currently leads generative AI and compliance-automation work at Softoo; previously led ML teams at Forbmax and Horizon Tech Services. Experienced across local and cloud LLM providers (Llama, DeepSeek, Azure OpenAI, OpenAI, GCP/Vertex AI, Groq) for secure, cost-efficient enterprise deployment.

EXPERIENCE

Softoo

May 2025 – Present | Islamabad, Pakistan

AI Lead Engineer — Generative AI, RAG Systems, AI Agents

- Lead agentic AI and RAG development for enterprise automation and decision-support platforms.
- Architect multi-agent frameworks capable of reasoning through multi-step tasks without a fixed script.
- Implement RAG pipelines with semantic search for accurate, context-aware knowledge retrieval.
- Deploy SQLCoder, Llama, and Code Llama on-premise for data-sensitive clients, reducing inference cost.
- Built an AI audit platform automating BCM, SAMA, and ISO compliance workflows end to end, including risk/control matrix generation, control testing, and reporting.

Forbmax (pvt) Ltd

May 2023 – May 2025 | Islamabad, Pakistan

AI Team Lead — Computer Vision, NLP, Data Science

- Led a team spanning computer vision, NLP, and general ML, setting technical direction and owning core architecture.
- Rebuilt model pipelines end to end, cutting time from raw data to deployment by ~40%.
- Fine-tuned transformer, YOLO, and vision-transformer models to client-specific data distributions.
- Built an active-learning annotation loop that cut manual labeling effort by ~70%.
- Drove adoption of API/microservices architecture for scalable AI product integration.

Horizon Tech Services

Jun 2020 – Apr 2023 | Islamabad, Pakistan

Sr. AI Engineer

- Owned end-to-end ML projects and mentored a team through data collection, training, and deployment.
- Built preprocessing pipelines that streamlined feature extraction and data augmentation.
- Improved object detection recognition and tracking accuracy through targeted fine-tuning.

Lampro Mellon IT Solutions Pvt. Ltd

Aug 2019 – May 2020 | Lahore, Pakistan

Python Developer

- Contributed to software components with a focus on verification and high-level design.
- Developed automated verification scripts to streamline testing and validation.

SELECTED PROJECTS

Enterprise Compliance AI Agent Platform

Sep 2026

AI audit platform covering BCM, SAMA, and ISO compliance work, including maturity assessments; plans and runs full audits end to end (risk/control matrix, control tests, reporting) with hybrid local/cloud LLMs and full decision traceability. Live at yanipro.ai.

Agentic AI, LangGraph, Risk & Control Automation, SAMA/ISO/BCM

ActPilot — AI Browser Agent

Aug 2026

Chrome extension that reads a page's content and screenshot, then fills fields, clicks, or opens links based on natural-language requests, with chart and PDF export. Runs on Azure OpenAI (vision) with automatic Groq fallback; built-in guardrail against unsafe auto-submission.

Chrome Extension (MV3), Azure OpenAI (Vision), Groq Fallback, FastAPI

Private Data Assistant — Text-to-SQL over Live Databases

Apr 2026

Natural-language querying over Postgres, MySQL, SQL Server, SQLite, or MongoDB with schema-aware table retrieval and chart generation. Five-layer, per-engine read-only execution guard (keyword blocking, forced rollback, timeouts, row caps).

Text-to-SQL, Multi-DB Adapters, Read-only SQL Guard, Fernet Encryption

Private Document Assistant — Agentic RAG Chatbot

Jan 2026

Private RAG chatbot over user documents (PDF/DOCX/scanned images with OCR), with per-user and per-chat isolation. Model decides per-turn whether to invoke retrieval; responses stream via SSE.

Agentic RAG, Azure OpenAI, SSE Streaming, Multi-tenant Isolation

Private Voice Assistant — Real-time AI Phone Helpline

Aug 2025

Runtime-configurable AI voice helpline over real phone calls (Twilio) or browser demo, sharing a Groq-powered STT → LLM → TTS pipeline tuned to 2–2.7s turn latency; auto-detects and responds in the caller's language.

Twilio Voice, Whisper STT, Orpheus TTS, Real-time Voice AI

WellB — AI-Powered Hotel Management System

Feb 2025

Hotel management chatbot combining local Llama/DeepSeek models with OpenAI GPT for natural-language booking management, priority-classified emergency alerts, and automated contact management.
RAG, Llama, OpenAI

IntelliReport — AI-Powered Report Generation Sep 2024
BERT-based topic extraction and extractive/abstractive summarization system with topic-wise summaries, automated highlighting, and a batch-processing API.
BERT, Text Summarization, Topic Extraction, FastAPI

Smart Media Monitoring Application Aug 2023
Media monitoring pipeline combining speech-to-text, sentiment analysis, summarization, topic modeling, facial recognition, and multi-language OCR (Urdu, English, Arabic) into one system.
Speech-to-Text, Facial Recognition, Multi-language OCR

Smart Surveillance System Dec 2022
Real-time monitoring across live and IP camera feeds: object detection, automatic number-plate recognition, geofencing, and video summarization.
Object Detection, ANPR, Geofencing

AAQAB — Satellite Imagery Intelligence Feb 2022
YOLO/ResNet-based object detection over high-resolution satellite imagery, with distributed GPU processing for large images and an automated PRSS analysis workflow.
YOLO, SAHI, Distributed GPU Processing

PUBLICATIONS

Meter Digit Recognition Via Faster R-CNN — Muhammad Waqar, M. Waris, et al., M. Yousaf Oct 2019
International Conference on Robotics and Automation in Industry (ICRAI) · DOI: 10.1109/ICRAI47710.2019.8967357

SKILLS

Generative AI & Agents	LLM Orchestration, Agentic Tool-Calling, RAG Pipelines, LangChain, LangGraph, Prompt Engineering, OpenAI SDK, Azure OpenAI, GCP/Vertex AI, Groq, On-Prem LLM Hosting
ML & Computer Vision	PyTorch, TensorFlow, Transformers, YOLO, scikit-learn, Keras, OpenCV
Voice & Speech AI	Groq Whisper STT, Azure Speech, Text-to-Speech, SeamlessM4T (Meta), Twilio Voice, Real-time Audio Pipelines
Data & Retrieval	PostgreSQL, MySQL, SQL Server, MongoDB, Text-to-SQL, SQLAlchemy, Alembic Migrations
Full-Stack & DevOps	FastAPI, React, TypeScript, Docker, JWT Auth, CI/CD (pytest/vitest), Vercel, Git
Leadership & Practice	Team Leadership, System Architecture, Security Engineering, Mentorship, Test-Driven Development

EDUCATION

University of Engineering and Technology (UET), Taxila — B.Sc. in Computer Engineering Oct 2015 – Aug 2019